GENERAL SPECIFICATION G3.23

Structural Steel Work

CONTENTS

G3.23 STRUCTURAL STEEL WORK

G3.23.1 Introduction

(a) This Part covers the requirements for the supply and erection of structural metalwork to designs by the S.O. or by the Contractor where such designs are proposed as part of the Permanent Works. This part applies to all such structural metal work including work in steel, stainless steel and aluminium.

G3.23.2 Contractor's Design Responsibility

(a) Except where the necessary details are shown on the S.O.'s drawings, the Contractor shall carry out the detailed design of connections between members (both bolted and welded) to the design data approved or provided by the S.O..

G3.23.3 Reference Standards

(a) Unless otherwise specified, structural metalwork shall comply with the following standards:

Standard	Subject
SS EN 1990	Basis of structural design
SS EN 1991	Actions of structures
SS EN 1993	Design of steel structures
BS EN 10025	Hot rolled products of structural steels
BS EN 10029	Tolerances on dimensions, shape and mass for hot-rolled steel plates 3 mm thick or above
BS EN 10113	Hot-rolled products in weldable fine grain structural steels
BS EN 10164	Steel products with improved deformation properties perpendicular to the surface of the product – Technical delivery requirements
BS EN 10210	Hot finished structural hollow sections of non-alloy and fine grain structural steels
BS EN 10219	Cold formed hollow sections of structural steel
BS EN 1090	Execution of structural steels structures – technical requirements
BS EN ISO 12944	Paints and varnishes – Corrosion protection of steel structures by protective paint systems
BS EN ISO 1461	Hot dip galvanised coating on fabricated iron and steel articles – specifications and test methods
BS EN ISO 3506-1	Mechanical properties of corrosion resistant steel fasteners part 1 – bolts, screws and studs

G3.23.4 Submissions by the Contractor

(a) The following submissions are required by this part of the Specification:

(i) Drawings

1) Drawings comprising general arrangements, general assembly, detailed shop and erection including identification of members.

2) The Contractor shall allow for the preparation of detail shop drawings. The shop drawings shall give complete information necessary for the fabrication of the component parts of the structure including the location, type, size and extent of all welds and bolt holes. All dimensions and levels including as-constructed holding down bolts positions affecting new steelwork shall be checked on site shall be checked by Contractor and incorporated into shop drawings, which shall be prepared from the working drawings with reference to the BIM model and other contract documents as required. Holes required in beams for air-conditioning and services and attachment of timber plates, etc. will be positioned by the Contractor at his own cost. However, approval shall be limited to matters relating to structural sufficiency only.

(b) Data

(i) Calculations for structural analysis and design where this is carried out by the Contractor, calculations for connection details, where requested by the S.O. and welding procedure sheets.

(c) Certificates

(i) Certificates comprising material tests, inspection certificates and qualification certificates compliance to BC1 requirement.

G3.23.5 Materials

(a) Structural Metalwork Generally

(i) Rolled structural steel sections shall be mild steel Grade S275, or, high yield steel to Grade 355; and

(ii) Where the use of proprietary designs of prefabricated building frames is proposed, the standards to which they are manufactured shall be no less rigorous than those stated above.

(b) Bolts and Nuts

(i) Steel bolts and nuts shall be high strength friction grip bolts, Grade 4.6, Grade 8.8 or Grade 10.9 as specified on the Drawings. Black metal washers to Form E shall be used unless otherwise specified;

(ii) High strength friction grip bolts shall be used in conjunction with approved proprietary load indicating washers;

(iii) All bolts and nuts shall be tightened to a snug tight fir unless otherwise stated; and

(iv) All nuts and washers shall comply with SS EN 1993 1-8.

(c) Anchor Bolts

(i) Cast-in anchors shall be used wherever possible, post installed anchors shall only be used with the approval of the S.O.. The complete anchorage system including anchor bolts, washers, locknuts, etc. shall be from one manufacturer and mixing of different components from different systems or manufacturers shall be prohibited. Adhesive anchors that have not been tested for creep or have failed creep test shall be prohibited from use. Test reports shall be submitted to the S.O. for acceptance prior to any installations. The complete anchorage systems shall be grade 316 stainless steel.

(ii) Detection devices shall be used to locate steel reinforcement before carrying out any drilling of holes for anchor installation. Holes shall be drilled and cleaned in accordance with manufacturer’s instructions. The minimum edge distance, spacing and embedment depth specified by the manufacturer shall be adhered to. The anchors shall be installed using the specified tools by the same manufacturer. Mixing of installation tools of different make by different manufacturers is prohibited. Where the manufacturer recommends the use of special tools for installation of anchors, such tools shall be used. The installation of anchors shall be carried out only by trained and certified applicators. Certificates of applicators shall be submitted to the S.O. for acceptance prior to any installation work.

(iii) When adhesive anchors are proposed, manufacturer’s information on the formation of the adhesive, creep characteristics, creep and other test reports shall be submitted for the S.O. for acceptance prior to use. The ultimate capacity of adhesive anchors shall be determined based on their long-term load carrying capacity appropriate to the design life span of the main structure, taking into consideration creep characteristics and temperature effects under sustained tensile loads.

(iv) Anchor shall be tested in accordance with ASTM E 488, ASTM E 1512 and ICC ES AC 58. Tests shall be carried out for 1 in every 50 anchors or less of the same type for each batch. For every anchor that fails, an additional 5 anchors shall be tested. If any of the 5 anchors do not meet the test criteria, tests shall be carried out on all the remaining anchors in the test batch.

(v) All anchors, nuts, washers to be sherardized to coating thickness of 43 micron.

(d) Welding Consumables

(i) Welding of structural aluminium, where permitted by the S.O., shall be in accordance with Section 7 of the Aluminium Association's Specification for Aluminium Structures and the American Welding Society D1.2-83 Structural Welding Code for Aluminium.

G3.23.6 Workmanship

(a) Design

(i) Where design of structural steel work is to be carried out by the Contractor, it shall comply with the relevant standards. Where design of structural aluminium is to be carried out by the Contractor it shall comply with the relevant standards. Designs shall be carried out by a Professional Engineer Singapore (Civil) and shall be submitted to the S.O. for approval prior to commencement of fabrication.

(b) Detailing

(i) The detailing and fabrication of structural metalwork shall be such as to minimise the need for site welding. Structures and components such as trusses, combined columns and beams shall be shop fabricated to form sub-assemblies of the largest practical size suitable for transportation, handling and erection. Connections between members at the erection stage shall generally be bolted.

(ii) Structural metalwork shall be so detailed and fabricated as to minimise the formation of pockets to hold condensation, water or dirt. Where such pockets are unavoidable, suitable drainage shall be provided.

(iii) Detailing shall be in accordance with the relevant standards.

(c) Detail Drawings and Welding Procedure Sheets

(i) The Contractor shall prepare detailed shop and erection drawings and welding procedure sheets for all structural metalwork in accordance with the relevant standards and these shall be submitted to the S.O. for approval before fabrication commences. The sequence of submissions shall match the order of fabrication and erection.

(ii) Shop drawings for fabrication shall show full details of the type of steel, member sizes and dimensions, weld details, welding sequences and any requirements for weld stress relieving.

(iii) Welding procedure sheets shall show all details of the welds including:

1) Welding method;
2) Current type, voltage and amperage;
3) Welding positions;
4) Preparation angles and methods;
5) Number of runs and their welding positions;
6) Type, size and class of electrode;
7) Shielding gas type and flow rate (where applicable);
8) Non-destructive testing methods and extent of coverage; and
9) Pre-heat, post-heat and stress relief methods, temperatures and the like.

(iv) Where substitution of alternative structural sections or modifications to the design or details is proposed, the Contractor shall prepare revised drawings, and supporting calculations if required, for consideration by the S.O.. Any such proposals by the Contractor shall not be put into effect unless the S.O. has given his approval.

(v) Erection drawings shall detail all connections, showing bolt types, tensions and sizes and site weld details as applicable.

(vi) Shop and erection drawings shall show all marks necessary to facilitate erection and the marks shown on the drawings shall be identical with those shown on the fabrications.

(d) Welding

(i) All welding carried out during fabrication or erection shall be in accordance with the requirements of the relevant reference standards and as shown on the approved detail drawings.

(ii) Welding shall be a metal arc process in accordance with the relevant standards, unless otherwise permitted by the S.O.. Details of the proposed weld procedures shall be submitted to the S.O. for approval at the same time as the detail drawings. All connections shall be welded in such a manner as to make the finished connections neat and smooth in appearance and suitable for painting. All slag shall be removed and any sharp projections shall be ground smooth.

(iii) Before welding of structural steel work is commenced either in the fabrication shop or on site, weld procedure tests shall be carried out in accordance with the relevant standards where directed by the S.O..

(iv) Equivalent weld procedure tests shall be carried out for other structural metalwork.

(v) All welders for structural steel work employed either in the fabrication shop or on site shall pass qualification tests relevant to the weld procedures in use in accordance with the relevant standards. Welders for other structural metalwork shall pass equivalent qualification tests. All welding works shall be carried out by certified welders tested in accordance with BS EN 287-1, all welding works shall comply with BS EN 1011-1. Welders shall have satisfactory evidence of having been engaged in welding for at least 9 months in the preceding 12-month period. If the work of any welders employed on the Contract is unsatisfactory, the Contractor shall carry out such further welder qualification tests as are necessary to demonstrate that the welders are proficient.

(vi) No site welding is allowed except where approved.

(vii) Welded connections between structural members shall have a 5mm or thickness of plate whichever being thicker, continuous fillet weld (CFW) unless noted otherwise.

(viii) The contractor shall carry out visual inspection on 100% of the weld witnessed by QP(S) or representative.

(ix) All joints shall be properly cleaned prior to welding.

(x) Welds shall be subjected to non-destructive testing by processes which may include but shall not necessarily be limited to radiographic, ultrasonic, magnetic particle, or dye penetration methods, depending on the type of weld and its position in the structure.

(xi) If any work shows defects or fails to comply with the requirements of the Drawings or Specification for any reason it shall be repaired or rejected, even though it may have been carried out by qualified welders using approved procedures.

(e) Fabrication Tolerances

(i) The general tolerance on all dimensions shall be in accordance with the relevant standards. Holes shall be aligned such that fasteners can be freely inserted through the members at right angles to the contact face. Where holes in members cannot be aligned without damaging or distorting the structure or reaming the holes (where permitted by the S.O.), the member or members shall be rejected.

(ii) A structural member shall not deviate from straightness (or from the specified shape) by more than 1/1000 of the length between lateral restraints or 3 mm whichever is greater in the case of compression members and beams.

(iii) A structural member shall not deviate from its intended length by more than 3mm.

(iv) Lengths of components shall be such that cumulative variations do not prejudice the accurate alignment of the completed structure.

(v) Where two metal surfaces are required to be in contact to effect a bearing or frictional contact, the surfaces shall be prepared so that at least 90% of the area is touching before any clamping force is applied.

(f) Dissimilar Metals

(i) Where dissimilar metals are used in proximity to structural members or their connections, contact between such metals shall be avoided unless the Contractor can demonstrate to the satisfaction of the S.O. that contact between the dissimilar metals will not lead to galvanic corrosion.

(ii) Contact between aluminium or aluminium alloy and galvanised mild steel will be permitted. For fixing aluminium to steel structures all fastenings such as bolts, nuts, washers and screws shall be zinc or cadmium plated.

(iii) Where zinc plated parts might otherwise become sacrificial anodes to the main structure, or where the electrolytic potential difference exceeds 250 mV, the parts shall be separated by an insulating medium of adequate strength.

(g) Identification of Members

(i) All members of assemblies shall be marked at the factory with distinguishing numbers, letters or marks corresponding to those on approved drawings or parts lists. Such marks if impressed before painting shall be clearly readable afterwards.

(ii) Any temporary bolts for field erection shall be clearly marked as such to distinguish them from bolts for permanent connections.

(h) Marking of Overhead Runway Beams

(i) The Contractor shall ensure that the safe working load, identification number and any limiting operating conditions are plainly and permanently marked on overhead runway beams to be clearly visible to the operators.

(i) Storage

(i) All structural elements and other materials shall be stored (whether at the place of fabrication or on Site) clear of the ground and shall be protected against deterioration from any cause. The Contractor shall agree the storage areas on the Site with the S.O. before any structural steel work is delivered.

(j) Erection

(i) The plant used for handling and erection shall be suitable and safe. The Contractor shall supply, maintain and afterwards remove temporary protection to workmen and materials below any structural metalwork under erection and shall provide walkways and ladders so that inspection of any part of the completed structure may be safely carried out.

(ii) Anchor bolts shall be properly located using templates or other approved method. After building in the anchor bolts, a sufficient time shall be allowed for concrete strength to develop before finally pulling down to shims and grouting.

(iii) Every part of the structural metalwork shall be positioned accurately in accordance with the dimensions on the approved drawings with a maximum tolerance of 5 mm except where the metalwork supports metal flooring, when the finished tolerances shall be such that differences in level between adjacent flooring sections or between flooring sections and adjacent building floors do not exceed 3 mm. Gaps between flooring sections shall not exceed 3 mm.

(iv) Where structural metalwork is to be grouted on or against a concrete or other similar surface and the design requires a uniform contact area under the base plate, or where directed by the S.O., the Contractor shall remove all shims or wedges and replace these with grout of the same constituents and consistency as the main grout. Such shims shall be initially positioned to aid removal. In the case of baseplates with large overhangs or with heavy bearing loads the positions of shims shall be agreed with the S.O. before the structure is erected on them.

(v) During erection, the work shall be securely bolted or otherwise fastened, and if necessary temporarily braced, to make adequate provision for all erection stresses and conditions, including those due to erection plant and its operation. No permanent bolting shall be done until proper alignment has been achieved.

(vi) The Contractor shall immediately report to the S.O. any errors in shop fabrication, and any deformations resulting from handling or transportation, which prevent proper assembly and fitting of parts. Methods to be used to correct such errors or deformations shall be agreed with the S.O. prior to beginning of any corrections.

(vii) The Contractor may use drift pins to bring several parts together but shall not use them in such a way as to cause distortion or damage.

(k) Bolted Connections

(i) When assembled, all joint surfaces, including those adjacent to bolt heads, nuts and washers, shall be free from burrs, dirt or other deleterious matter or defects preventing proper seating of the parts.

(ii) For structural steelwork, the Contractor shall make bolted connections in accordance with the details shown on the shop or erection drawings and with the relevant standards. The bolts used shall be in accordance with this specification, as detailed on the Drawings. When the bolt head or nut is in contact with a surface which is inclined at more than 3E from a plane at right angles to the bolt axis, a taper washer shall be placed to achieve satisfactory bearing. The bolt length shall be such that each bolt shall project through the nut not less than one thread plus the thread run-out and not more than five threads when fully tightened; the length of the threaded portion shall be such that there shall be at least two threads under the nut after tightening. Additionally, foundation bolts shall have a minimum thread length equal to the bolt diameter plus 100 mm.

(iii) The Contractor shall ensure that the facing surfaces (i.e. those surfaces required to transmit friction in conjunction with high strength friction grip bolts) are kept free of contamination by grease, rust, moisture, dust and the like. If the surfaces are primed after fabrication, the primer shall be suitable for use in such connections and shall be protected from damage during transit or erection. Should contamination occur, it shall be remedied by appropriate means, such as:

1) On coated surfaces, with a degreasing agent followed by washing and drying; and

2) On uncoated surfaces, as above, or by blast cleaning or wire brushing to clean steel.

(iv) These remedial measures shall be taken immediately before the surfaces are bolted together.

(v) High strength friction grip bolts shall be used in accordance with the relevant standards, using approved load-indicating washers to check bolt tension. They shall be tightened in accordance with the manufacturers' recommendations and the tension shall be rechecked not less than three hours after first tightening. The bolts shall then be re-tightened to the initial load, all to the approval of the S.O..

(vi) For other structural metalwork bolted connections shall be formed in accordance with relevant standards.

G3.23.7 Testing

(a) The S.O. may inspect structural metalwork during fabrication and witness testing at the fabrication works. The Contractor shall give the S.O. adequate notice of such operations in order that the necessary arrangements may be made.

(b) The Contractor shall arrange for the S.O. and his authorised representatives to have access to the fabrication works at all reasonable times and to be provided with all the necessary facilities and test equipment to carry out such inspections.

(c) An Independent Testing Agency (ITA) to provide inspection and testing services to verify that the Contractor's inspection, testing and quality control work is being properly performed, and to make random spot checks as deemed appropriate.

(d) Materials and workmanship shall be subject to inspection and testing at the shop and/or at the field by the S.O. and/or the ITA. Such inspection and testing shall not relieve the Contractor of his responsibility to provide his own inspection, testing and quality control as necessary to furnish the materials and workmanship in accordance with the requirements of the Contract.

(e) Before any steel overhead runway beam is taken into use for the first time, the Contractor shall submit to the S.O. a signed certificate from the manufacturer in accordance with SS EN 1993.

(f) Testing of Structural Steel shall include the following:

(i) Chemical characteristics test;

(ii) Tensile / elongation test; and

(iii) Charpy impact test.

(g) Frequency of testing shall be 1 sample for each size of steel member from each source of supply per every 100 Ton or part thereof received at site. The test frequency shall be followed unless otherwise specified and agreed by S.O..

(h) Testing of Welds

All welded connections, comprising fillet welds or full penetration butt welds, shall be subjected to non-destructive tests to be conducted by accredited laboratories to prove the soundness of welds, the testing of welds shall be at the discretion of the S.O.. The contractor is required to engage an independent accredited testing agency approved by the S.O. as the welding inspector. The welding inspector shall propose an inspection and testing plan for the quality control of the welds. The proposed inspection and testing plan shall include all the necessary weld tests and its frequency and shall be approved by the S.O. before implementation.

(i) Weld measurement shall be performed for 15% of all welds on a random basis.

(ii) Magnetic particle testing to BS EN ISO 17638 shall be performed for a minimum of:

1) 100% of all fillet welds done at site;

2) 50% of all fillet welds done at factory; and

3) 100% of all partial penetration welds.

(iii) Ultrasonic testing in accordance with BS EN ISO 17640 shall be performed for a minimum of:

1) 100% of all full penetration welds; and

2) 20% of all partial penetration welds.

G3.23.8 Corrosion Protection

(a) Protective Paint

The structural steel works and all ancillary structural steel works including all connections which are not encased in concrete shall be coated with the following system of protective paints or an approved equivalent in accordance with the relevant standard:

Life to First Maintenance	:	10 years
Surface Preparation	:	Abrasive Blast Cleaned to SIS 055900 Sa 2½ or SSPC-SP10
Primer	:	Inorganic zinc silicate - 75 microns dry film thickness
Barrier Coat	:	2-pack high build micaceous iron oxide polyamide epoxy - 125 microns dry film thickness
Finish	:	2-pack polyurethane - 125 microns dry film thickness

(b) Galvanising

(i) All hot rolled sections, plates, flats, bars and hollow sections which are specified to be galvanised are to be hot dip galvanised with a minimum coating thickness of 85 microns. After hot dip galvanising, all the steel work shall be dipped in a 0.2 percent chromic acid solution.

(ii) All black bolts, mild steel fasteners and fixings which are specified to be galvanised are to be hot dip galvanised to achieve a minimum average coating thickness of 52 microns, and a minimum individual coating thickness of 43 microns. High strength friction grip bolts, hardened washers and nuts shall not be hot dip galvanised.

G3.23.9 Bedding and Grouting

(a) Where steel work is supported by concrete, masonry or like material, it shall be set up on packing or wedges to facilitate alignment and permit subsequent grouting. Such packs, if permanent, shall be of grout of similar strength to the permanent grout. All other packs shall be removed before completion of grouting.

(b) Grout

(i) Cement Based Grouts

1) Cement based grouts used for bedding steel bases or bearing plates on concrete foundations shall be of one of the following forms:

- Neat Portland cement of a thickness not exceeding 25mm. - The grout shall be mixed as thickly as possible consistent with fluidity and shall be poured under a suitable head so that the space is completely filled.

- Fluid Portland cement mortar of a thickness between 25mm and 50mm. The mortar shall not be leaner than 1:1 cement to fine aggregate and shall be mixed as thickly as possible consistent with fluidity. The mortar shall then be poured under a suitable head and tamped until the space has been completely filled.

2) Dry as possible Portland cement mortar of thickness not less than 50mm. The mortar shall not be leaner than 1:2 cement to fine aggregate and shall be consolidated by thoroughly ramming with a suitable blunt rammer against properly fixed supports until the space has been completely filled.

(ii) Proprietary Grouts

1) Where specified by the S.O., the following proprietary high-strength non-shrink grouts may be used:

- Mortar or fine concrete containing suitable admixtures, including the use of expanding additives to avoid shrinkage;
- Expanding grout; and
- Resin based grout.

(iii) Strength of Grouts

1) All grouts around foundation bolts and under column base plates shall have a minimum compressive strength at 28 days (based on 50mm grout cubes) of not less than $60N/mm^2$.

(c) The Contractor shall be responsible for the grouting of all base plates, bearing plates and for end connections where shown on drawings. Grout shall be neatly finished and tapered at the edges by steel trowelling.

G3.23.10 Fireproofing Materials

(a) The fireproofing is to be spray applied to all the structural members not encased in concrete and under internal environment only. Where shown on the Drawings an approved fireproofing is to be applied to structural steel members which are unencased and are exposed to the elements.

(b) Material containing asbestos or mineral wool will not be permitted.

(c) Spray-on fireproofing shall not erode, delaminate, flake or dust when subject to an air stream with a velocity of at least 5.3m/s for 24 hours. The Contractor shall apply a sealer to prevent erosion, flaking and dusting. The Contractor shall submit test data and application procedures of spray-on fireproofing and of sealer.

(d) The Contractor shall demonstrate that spray-on fireproofing material inhibits corrosion of steel when applied to unpainted steel when tested in accordance with ASTM E-937. All fireproofing materials shall be certified and approved under SCDF Fire Code requirements and regulations.

END OF SECTION